and gravitational forces, this view was augmented to include the value of each field—the strength of each field, that is—and the rate of change of each field's value, at every location in space. But by the 1930s, the uncertainty principle dismantled this conception of reality by showing that you can't ever know both the position and the velocity of a particle; you can't ever know both the value of a field at some location in space and how quickly the field value is changing. Quantum uncertainty forbids it.

As we discussed in the last chapter, this quantum uncertainty ensures that the microworld is a turbulent and jittery realm. Earlier, we focused on uncertainty-induced quantum jitters for the inflaton field, but quantum uncertainty applies to all fields. The electromagnetic field, the strong and weak nuclear force fields, and the gravitational field are all subject to frenzied quantum jitters on microscopic scales. In fact, these field jitters exist even in space you'd normally think of as empty, space that would seem to contain no matter and no fields. This is an idea of critical importance, but if you haven't encountered it previously, it's natural to be puzzled. If a region of space contains nothing—if it's a vacuum—doesn't that mean there's nothing to jitter? Well, we've already learned that the concept of nothingness is subtle. Just think of the Higgs ocean that modern theory claims to permeate empty space. The quantum jitters I'm now referring to serve only to make the notion of "nothing" subtler still. Here's what I mean.

In prequantum (and pre-Higgs) physics, we'd declare a region of space completely empty if it contained no particles and the value of every field was uniformly zero.[30] Let's now think about this classical notion of emptiness in light of the quantum uncertainty principle. If a field were to have and maintain a vanishing value, we would know its value—zero— and also the rate of change of its value—zero, too. But according to the uncertainty principle, it's impossible for both these properties to be definite. Instead, if a field has a definite value at some moment, zero in the case at hand, the uncertainty principle tells us that its rate of change is completely random. And a random rate of change means that in subsequent moments the field's value will randomly jitter up and down, even in what we normally think of as completely empty space. So the intuitive notion of emptiness, one in which all fields have